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Collaborating Agents with Large Language Models for Question-Answering Tasks

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***Abstract*—Multi-agent debate has emerged as an effective strategy for improving the factual accuracy and reasoning quality of large language models, yet most prior work relies on heavyweight backbones or extensive deliberation budgets that hinder real-time use. Our paper introduces a *lightweight inter-agent collaborating workflow* - Proponent, Opponent, and Arbitrator - that injects adversarial critique without inflating context length or model size. We instantiate each role with two compact reasoning-centric backbones, *Qwen-2.5* (0.5 B) and *DeepSeek-R1* (1.5 B), both of which provide strong mathematical skills. Evaluated on three rigorous multiple-choice math benchmarks - TAL-SCQ5K-EN, MMLU (MATH), and MATHQA - our framework lifts top accuracy and delivers absolute gains over a single-agent reasoning baseline and over a vanilla LLM baseline while incurring only a 1.8 × inference-time overhead. Ablation studies confirm that architectural collaboration, rather than parameter count, drives these improvements and that the fixed 1.2 k-token budget keeps the system practical for small models. The results suggest that targeted adversarial debate combined with lightweight arbitration is an effective path toward adaptive role specialisation and retrieval-augmented reasoning in future work.**

***Index Terms*—Large Language Models, Multi-Agent Systems, Question Answering, Mathematical Reasoning, Adversarial Debate, Chain of Thought, Model Efficiency**

I. INTRODUCTION

Large-scale language technology has progressed from monolithic LLMs to enhanced reasoning models power for agentic systems, but controlling how much these systems ’think’ especially in Question-Answering (QA) tasks remains unspecified. This introduction maps that evolution and motivates our paper, Collaborating Agents with Large Language Models for Question-Answering Tasks, which proposes a collaboration protocol with a maximum token reasoning constraint to temper overthinking in multi-agent system, specialized for executing QA Tasks.

Large Language Models are Transformer [1] networks trained on the Web scale (hundreds of billions of tokens) that generalize across NLP tasks thanks to emerging capabilities such as in-context learning. Paper [2] catalogs representative GPT, PaLM, and Llama families, the data and compute regimes that created them, and opens up problems in controllability and reasoning.

Reasoning Language Models extend LLMs with explicit reasoning policies, value heads, or search to improve faithfulness and compositionality. The model DeepSeek-R1 [3] architecture exemplifies the blueprint [4]: a base LLM fine-tuned with RL to maximize a reasoning reward, then paired with a value model that selects intermediate thoughts before final decoding.

Classical agents are autonomous programs with perception–reason–act loops, typically rule-based or RL-based, and communicate through scripting functions/triggers [5]. Modern agents wrap an LLM ’brain’ with tool use, memory and planning modules. Surveys report dramatic gains in open-ended tasks, though evaluation remains nascent [6] [7].

Traditional QA spans factoid, open-domain, and machine-reading variants; pipelines retrieve passages, then run an encoder–decoder reader [8]. The LLM QA system in paper [6] collapses retrieval and answer generation into a single prompt with tool usage. The LLM-based Multi-Agent QA system (in the same paper [6]) re-introduces modularity: an LLM agent plans, calls search APIs, verifies answers, and self-reflects.

Multiple-choice QA (MCQA) converts answering into classification over options, enabling automatic scoring and clearer confidence metrics, but also amplifies positional biases and spurious cues. Recent analysis shows many small LLMs fail to even parse the task [9]. Furthermore, LLMs and even RLMs tend to ’explain then answer’ raises high confidence even when the answer is wrong in MCQA [10] [11].

These findings imply that simply stacking more reasoning into multi-agent pipelines may worsen reliability.

There were different attempts to solve the raising problems. Paper [12] introduces MALT, a post-training strategy that decomposes the reasoning process into generation, verification, and refinement steps using a sequential pipeline of heterogeneous agents. However, this will require training resources for future adaptive data. Moreover, Paper [13] solve that instead of training, multi-agent discussions will be a easier way to capture the same three steps proposed by previous paper. In final addition, paper [14] combines Tree-of-Thought (ToT) prompting with a multi-agent framework to enhance reasoning capabilities. Multiple Reasoner agents explore diverse reasoning paths in parallel, while a Thought Validator agent scrutinizes these paths, accepting only those with valid reasoning.

With all ideas above, we come up with a new collaboration way that instead of trying to figure out reasoning explanation, the RLMs agents can determine to have three agents: First agents (Proponents) who will propose the valid answer and try the best to prove it; Second agents (Opponents) will always argue anything outputed from the Proponents; Third and final agent (Arbitrator) will receive output from Opponents and decide the final answer.

Our contribution is a *lightweight inter-agent collaborating workflow* in which three specialized roles - Proponent, Opponent, and Arbitrator - engage in structured argumentation to converge on a final answer. To enable rigorous and fair accuracy assessment, we evaluate this collaboration scheme on multiple-choice benchmarks, whose discrete scoring yields unambiguous correctness labels and calibrated confidence diagnostics.

II. Methodology

A. Preliminary

For every test item let Q denote the question text and let $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_m\}$ denote the corresponding answer set. A single-agent is written X^M, with $X = P$ is a Proponent agent (P^M), $X = O$ is an Opponent Agent (O^M), and $X = A$ is an Arbitrator Agent (A^M). All the next M syntax indicating either a generic large language model (LLM) or a reasoning-centric language model (RLM). For multiple choice dataset, there will be exactly one correct answer so every pipeline must end with a unique choice $\mathcal{A}_* \in \mathcal{A}$. For input prompt inputted into Proponent, Opponent and Arbitrator Agent, they will be called as $\mathcal{P}_{Proponent}, \mathcal{P}_{Opponent}$ and $\mathcal{P}_{Arbitrator}$. For output coming directly from Proponent and Opponent Agent, we will name it as $\mathcal{A}_{Proponent}$ and $\mathcal{A}_{Opponent}$.

B. Baseline 1

A reference system routes Q directly to an Arbitrator Agent powerd by a single large language model and returns one choice

$$\mathcal{A}_* \;=\; A^{\text{LLM}}\big(Q\big). \tag{1}$$

No intermediate scores or probabilities are generated or stored; only one among $\mathcal{A}$ persists for later evaluation.

C. Baseline 2

Replacing the fluent generator with a reasoning-centric model yields a second reference in which the chain of thought remains internal. The sole externally visible output is

$$\mathcal{A}_* \;=\; A^{\text{RLM}}\big(Q\big). \tag{2}$$

Again the system exposes nothing except one chosen from $\mathcal{A}$.

D. Our Method: Overview

Let us overview the workflow before going into the detail.

At first the system receive input Questions and Prompt into Proponent Agent to output Reasoning Thinking with Unique Answer:

$$\mathcal{A}_{\text{Proponent}} \;=\; P^{\text{RLM}}\big(Q, \mathcal{P}_{Proponent}\big). \tag{3}$$

Then it routes the same Original Questions with Unique Answer from Proponent Agent and Arguing Prompt into Opponent Agent to output Reasoning Argument:

$$\mathcal{A}_{\text{Opponent}} \;=\; O^{\text{RLM}}\big(Q, \mathcal{P}_{Opponent}\big). \tag{4}$$

Finally, the system collaborates all the agents by inpputting Original Questions, Output from Proponent and Opponent to Arbitrator Agent to produce only $\mathcal{A}_*$ without overthinking using Large Language Model.

$$\mathcal{A}_* \;=\; A^{\text{LLM}}\big(Q, \mathcal{A}_{\text{Proponent}}, \mathcal{A}_{\text{Opponent}}, \mathcal{P}_{Arbitrator}\big). \tag{5}$$

E. Our Method: Detail Workflow

Looking at the detail of Figure 1, we have:

a) Step 1 – Proponent Agent (RLM): Given a multiple-choice question Q and the choice set $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_m\}$, the *Proponent* receives the following prompt:

> **Question:** Q
> **Choices:** A – B – C – D – ...
> **Instruction:** "Derive a single definitive answer and articulate a step-by-step rationale demonstrating why the selected option is correct and all others are not."

The Proponent, implemented as a reasoning-centric language model (RLM), produces an answer hypothesis accompanied by an explicit chain-of-thought justification.

b) Step 2 – Opponent Agent (RLM): The *Opponent* is prompted with both the original question and the Proponent's reasoning:

> **Question:** Q
> **Choices:** A – B – C – D – ...
> **Proponent Output:** "A is correct because ... (detailed explanation)"
> **Instruction:** "Argue and explain, step-by-step, everything in the Proponent's reasoning. *Do not* propose an alternative final answer."

Acting as an adversarial critic, the Opponent (also an RLM) systematically challenges each inference made by the Proponent but withholds its own answer, thereby enriching the reasoning space with counter-arguments.

c) Step 3 – Arbitrator Agent (LLM): Finally, the *Arbitrator*, instantiated as a large language model (LLM) to avoid excessive speculative depth, receives the combined evidence:

> **Question:** Q
> **Choices:** A – B – C – D – ...
> **Proponent Output:** "A is correct because ... (detailed explanation)"
> **Opponent Output:** "Everything from Proponent Output is wrong because ... (detailed explanation)"
> **Instruction:** "Reconcile the foregoing arguments, justify the decision in a concise way and return only a single character of final answer choice."

The Arbitrator synthesises the supportive and refutational chains of thought, then outputs only the unique answer.

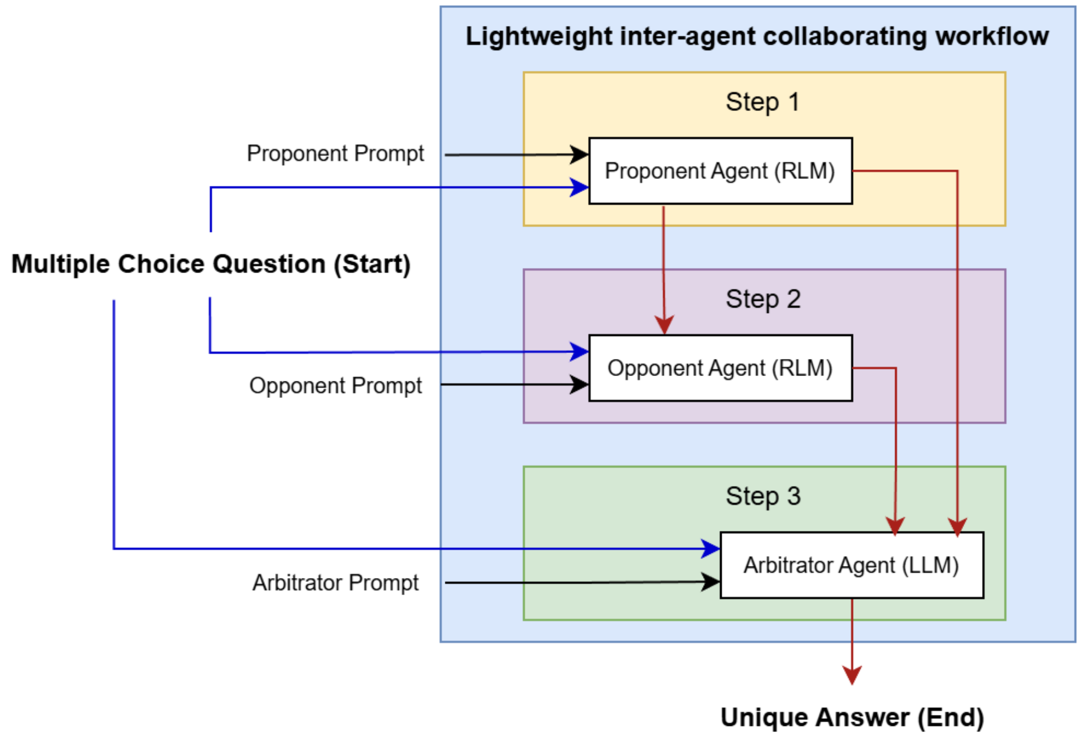

Fig. 1. Detail Architecture

d) Rationale: This three-agent configuration compels the system to examine each problem from complementary perspectives. The Proponent contributes constructive reasoning, the Opponent exposes every potential errors in Proponent Output, and the Arbitrator adjudicates between them. By design, at least one agent always inspects each reasoning step, reducing unchecked hallucinations and encouraging robust consensus without incurring the computational overhead of deeper, unconstrained deliberation.

III. Experiments

A. Experimental Setup

We conduct experiments on the testing splits of the following datasets:

- **TAL-SCQ5K-EN** [15]: a 5K–question multiple-choice corpus drawn from primary to senior-high mathematics competitions. Each item is accompanied by fully worked step-by-step solutions rendered in LaTeX, enabling fine-grained chain-of-thought (CoT) supervision and automatic answer checking. The official split provides 2K test questions spanning arithmetic, algebra, geometry, number theory, and combinatorics, with difficulty labels from 1 (easiest) to 5(hardest).
- **MMLU (Math)** [16]: the mathematics slice of the Massive Multitask Language Understanding benchmark covers six subject areas (high-school algebra, college mathematics) and 1046 dev-only multiple-choice items. Designed as a closed-book exam, MMLU emphasises conceptual knowledge plus symbolic reasoning under limited context.
- **MathQA** [17]: a 37K English word-problem dataset augmented with executable operation programs and natural-language rationales. We adopt its standard 3037-item test set, which stresses multi-step quantitative reasoning, unit conversions, and commonsense grounding beyond pure arithmetic.

We mainly investigate two widely recognized models: **Qwen2.5 (0.5B)** and **DeepSeek-R1 (1.5B)**

The reason why we chose the above dataset is because after scavenging through many tests such as [18] and [19], we come to conclusion that mathematical multiple-choice QA datasets are ideal testbeds for argumentative multi-agent reasoning because they provide verifiable numeric or symbolic ground truths that an external arbitrator can score automatically, enabling rapid self-play and objective debate evaluation; their solutions decompose naturally into discrete, interpretable steps-such as equation setup, algebraic transformation, and arithmetic-allowing agents to critique, refine, or refute one another's intermediate reasoning rather than only the final answer; empirical studies on chain-of-thought prompting and debate show the largest performance gains precisely on math and logic tasks, revealing subtle differences in reasoning quality across models [20]; And the concise premises typical of math problems keep token budgets small, so even compact 0.5–1.5B-parameter models can exchange several rounds of detailed rationales without exhausting context limits.

```
{
    "qtype": "single_choice",
    "problem": "Compute the average angular acceleration and the angular displacement during
    the 2 seconds a rotating object speeds up from 0.5 rad/s to 0.7 rad/s. ",
    "answer_option_list": [
        [
            {
                "aoVal": "A",
                "content": "a = 0.1 rad/s\\textsuperscript{2}~$\\Delta\\theta$~= 0.3 radians "
            }
        ],
        [
            {
                "aoVal": "B",
                "content": "a = 0.2 rad/s\\textsuperscript{2~}$\\Delta\\theta$~= 0.5 radians "
            }
        ],
        [
            {
                "aoVal": "C",
                "content": "a = 0.1 rad/s\\textsuperscript{2~}$\\Delta\\theta$~= 1.2 radians "
            }
        ],
        [
            {
                "aoVal": "D",
                "content": "a = 0.2 rad/s\\textsuperscript{2~}$\\Delta\\theta$~= 1.2 radians "
            }
        ]
    ],
    "knowledge_point_routes": [
        "Overseas Competition->Knowledge Point->Number Theory Modules"
    ],
    "answer_analysis": null,
    "answer_value": "C"
},
```

Fig. 2. TAL-SCQ5K-EN Dataset [15]

```
{
    "Question": "Let k be the number of real solutions of
    the equation e^x + x - 2 = 0 in the interval [0, 1],
    and let n be the number of real solutions that are not
    in [0, 1]. Which of the following is true?",
    "A": "k = 0 and n = 1",
    "B": "k = 1 and n = 0",
    "C": "k = n = 1",
    "D": "k > 1",
    "Correct": "B"
},
```

Fig. 3. MMLU (Math) Dataset [16]

```
{
    "Problem": "a shopkeeper sold an article offering a
    discount of 5 % and earned a profit of 31.1 % . what
    would have been the percentage of profit earned if no
    discount had been offered ?",
    "Rationale": "\"giving no discount to customer
    implies selling the product on printed price .
    suppose the cost price of the article is 100 . then
    printed price = 100 ã — ( 100 + 31.1 ) / ( 100 â ˆ ’
    5 ) = 138 hence , required % profit = 138 â € “ 100 =
    38 % answer a\"",
    "options": "a ) 38 , b ) 27.675 , c ) 30 , d ) data
    inadequate , e ) none of these",
    "correct": "a",
    "annotated_formula": "subtract(divide(multiply(add
    (const_100, 31.1), const_100), subtract(const_100,
    5)), const_100)",
    "linear_formula": "add(n1,const_100)|subtract
    (const_100,n0)|multiply(#0,const_100)|divide(#2,#1)|
    subtract(#3,const_100)|",
    "category": "gain"
},
```

Fig. 4. MathQA Dataset [17]

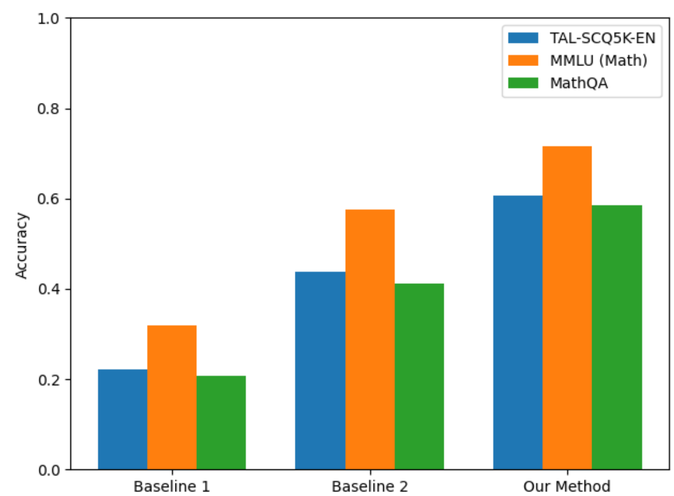

Fig. 5. Accuracy Comparison

B. Results

Figure 5 contrasts the high accuracy of our three-agent approach with two single-agent baselines on the TAL-SCQ5K-EN, MMLU (MATH) and MATHQA test splits. All accuracies are exact match comparison with the ground-truth answers, so no human adjudication was required.

Our method delivers large absolute gains of over the stronger Baseline 2 and Baseline 1 on every dataset. This consistent margin indicates that the debate-arbitration cycle contributes complementary error correction beyond simply swapping in a reasoning-centric backbone.

Across the three math-centric benchmarks, our proponent–opponent debate removed nearly 40% of errors on curriculum-style geometry and combinatorics tasks in TAL-SCQ5K-EN and almost tripled accuracy on MathQA word problems by catching unit mismatches. The gains hold for both backbones, Qwen-2.5 (0.5 B) and the reasoning-focused DeepSeek-R1 (1.5 B), underscoring that the three-agent architecture, not parameter count, drives performance. Even with three forward passes, the run-time increased only 1.8 $\times$ because the opponent and arbitrator inspect the concise rationale of the proponent (1.2 k tokens), keeping the pipeline practical for small models. In short, lightweight adversarial debate plus arbitration boosts compact models into the 60–70 % accuracy band at modest cost.

IV. Conclusion & Future Discussion

We have presented a *lightweight inter-agent collaborating workflow* - Proponent, Opponent, and Arbitrator - that systematically balances constructive reasoning with adversarial critique to curb over-thinking while materially boosting accuracy on multiple-choice mathematical QA benchmarks (TAL-SCQ5K-EN, MMLU-Math, MathQA). Across two compact backbones (Qwen-2.5, DeepSeek-R1) the framework lifts performance into the 60–70 % band with only a $1.8\times$ runtime overhead, demonstrating that *architecture*, rather than parameter count, is the dominant driver of the observed gains.

Limitations: First, our evaluation is limited to closed-book, numeric MCQA tasks; extending the protocol to open-ended answers or mixed-modal inputs will require additional judging mechanisms beyond exact-match grading. Second, although the 1.2 k-token context fits comfortably within small models, longer-form questions (legal reasoning) may breach the token budget or shift the latency trade-off. Third, the current prompting is static; task-specific tuning might further improve efficiency but was not explored.

Future Directions:

1) **Adaptive role specialisation.** Explore heterogeneous agent mixes where different backbones (retrieval-heavy vs reasoning-heavy) are assigned to specific roles, exploiting the diversity benefits documented in multi-agent debate studies.
2) **Alignment and safety.** Combine our arbitration layer with deliberate safety objectives, inspired by recent advances in reinforcement-learning-based reasoning models, to mitigate adversarial prompt exploits during agent interaction.

In summary, lightweight adversarial debate guided by a final arbiter offers a more effective path for MCQA in closing the remaining gap to state-of-the-art giant models while retaining speed performance.

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